THE CYCLE OF A FLAT-IN-HIGH-DIMENSIONS CLOSED SUBSCHEME IS A RELATIVE CYCLE

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Allowed standard prerequisites (editorial).

- Cycle family associated to a closed subscheme (relative-cycles chapter, Example example-family-associated-closed, canonical tag 0H4S). cited at statement line 1148. Defines the construction [Z/X/S]_r whose output the lemma asserts to be a relative r-cycle.
- Specialization compatibility of the module-associated family (relative-cycles chapter, Lemma lemma-family-associated-module-specialization, canonical tag 0H56). cited at proof lines 1155-1157. Applied to F = (Z -> X)_* O_Z to conclude the family satisfies the specialization condition.

Source reference. Original reference 0H4S: sources/stacks-chow/tags.txt, source lines 21362–21362. Cited at statement line 1148: defines the construction [Z/X/S]_r.

Source reference. Original lemma 0H56: sources/stacks-chow/tags.txt, source lines 21376–21376. Cited at proof lines 1155-1157: applied to F = (Z -> X)_* O_Z.

Lemma 0.2. *In the situation of Example 0H4S assume S is locally Noetherian and Z is flat over S in dimensions $\geq r$. Then $[Z/X/S]_r$ is a relative r-cycle on X/S.*

Proof. The assumption means that $\mathcal{O}_Z$ is flat over S in dimensions $\geq r$. Thus applying Lemma 0H56 with $\mathcal{F} = (Z \to X)_*\mathcal{O}_Z$ we conclude. □

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